K-FLATNESS IS A TWO OUT OF THREE PROPERTY FOR A DISTINGUISHED TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: it names both inputs, the cited reformulation of K-flatness and the two out of three property for acyclic complexes in a distinguished triangle.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules. Same chapter, the definition of K-flatness through acyclicity of total tensor products, and its reformulation as exactness of the derived tensor product functor. Supplies the property shown to be two out of three, in the reformulation supplied by the cited lemma.
- Distinguished triangle in the homotopy category K(Mod(O)). Same chapter, distinguished triangles in the homotopy category of complexes of modules over a ringed site. Supplies the triangle whose two out of three acyclicity statement is transferred to the two out of three exactness statement.
- Two out of three for acyclicity in a distinguished triangle. Same chapter, the long exact cohomology sequence of a distinguished triangle, which shows that if two of the three complexes are acyclic so is the third. Supplies the standard fact combined with the cited lemma to reach the conclusion.

Source reference. Original lemma 06YM: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2969–2969. Same-chapter navigation pointer invoked at proof line 3090, where the author reduces K-flatness to exactness of the derived tensor product functor. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{K}_1^\bullet, \mathcal{K}_2^\bullet, \mathcal{K}_3^\bullet)$ be a distinguished triangle in $K(Mod(\mathcal{O}))$. If two out of three of $\mathcal{K}_i^\bullet$ are K-flat, so is the third.*

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Proof. Follows from Lemma 06YM and the fact that in a distinguished triangle in $K(\mathit{Mod}(\mathcal{O}))$ if two out of three are acyclic, so is the third. □